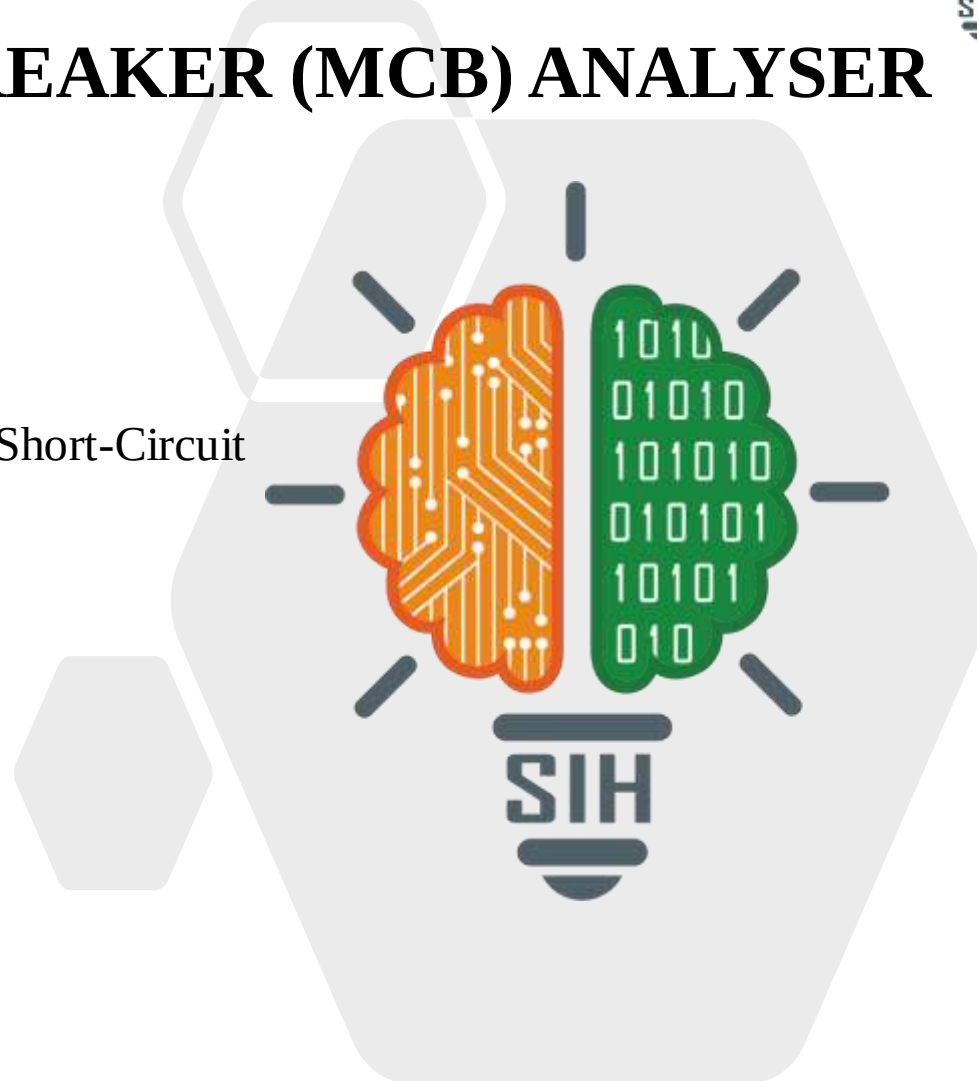


SMART INDIA HACKATHON 2026



MINIATURE CIRCUIT BREAKER (MCB) ANALYSER

- **Problem Statement ID** – 26029
- **Problem Statement Title**- Automated High-Current Short-Circuit Test System for IEC 60898-1:2015 MCB Compliance.
- **Theme**- Smart Automation
- **PS Category**- Hardware
- **Team ID**-
- **Team Name** - VOLTRA



MCB ANALYSER is an automated MCB testing system that measures current, detects tripping, calculates trip time, and logs the test results for IEC 60898-1:2015-based demonstration.

Problem Statement :

The problem is to develop an **automated system** that can perform and **monitor these MCB short-circuit tests accurately, safely, and repeatedly**. Existing MCB testing methods can involve **complex manual setup and repeated wiring**, making the testing process time-consuming and **difficult to maintain consistently**.

Proposed Solution :

A smart test system that **automates fault generation and MCB trip detection** with controlled switching. It monitors **voltage, current, waveforms, and response time** during the test. The system automatically **isolates the supply after fault detection** and records the test results.

Idea :

Automated MCB Testing : Automatically applies controlled load conditions and monitors MCB operation.
Real-Time Measurement : Measures voltage and current using sensors and detects the trip condition.
IEC 60898-1:2015 Reference : Demonstrates the MCB testing and trip-time monitoring concept based on the standard.

Uniqueness :

Automated fault application and **MCB tripping without manual intervention**.
Modular R/XL selection with **real-time monitoring** of voltage, current, waveform, and trip response time.
Integrated safety control with automatic shutdown during abnormal conditions, reducing operator involvement.

Technologies Involved :

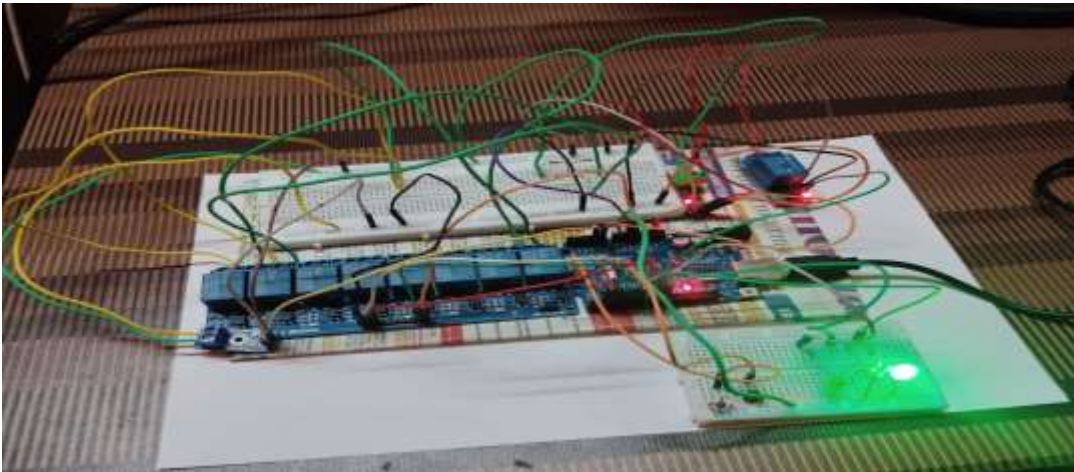
Arduino UNO : Main controller for automated test sequencing and relay control.

Relay-based switching : Automated R/XL selection, fault generation, and MCB trip simulation

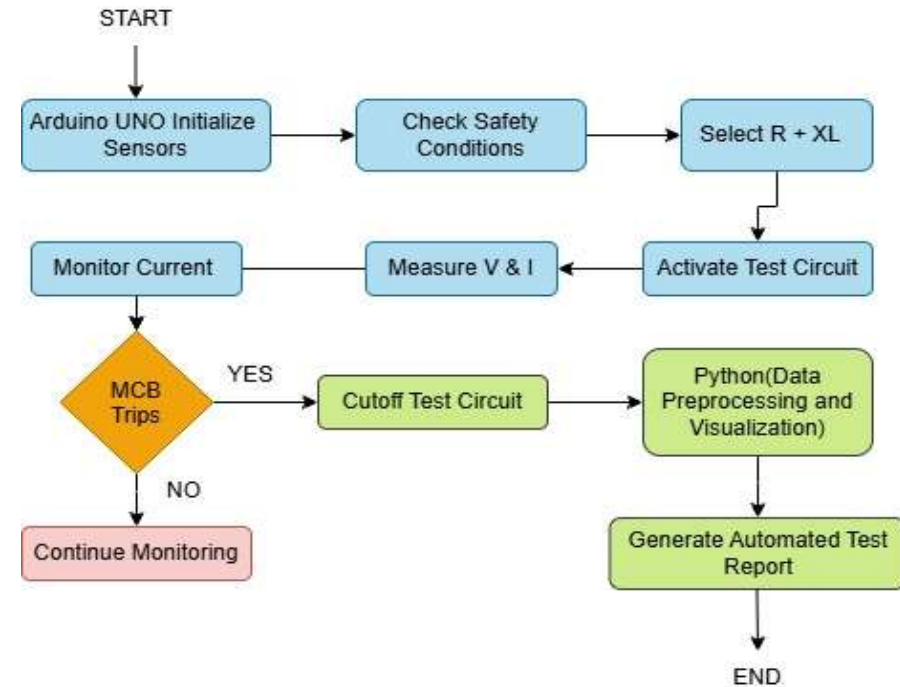
Proteus 8 : Circuit design and simulation of the complete prototype.

Python : Data analysis, visualization, and test-result processing.

Prototype :



System Architecture :



GitHub Link : https://github.com/ShaikSadh912/SIH2026_VOLTRA.git

Video Link : <https://youtu.be/iz6N5fo75uA?si=RvOdJQqPJj75Z5D6>

Product Status : 80% of work has been completed and remaining work is still in progress

Technical Feasibility :

1. **Scalable** : Modular design supports **future expansion to higher-current testing**.
2. **Reliable** : **Automated switching** and real-time sensing ensure **consistent operation**.
3. **Resources** : Required components and development resources are **readily available and cost-effective**.
4. **Feasible** : Embedded control enables **monitoring, protection, and data acquisition**.

Challenges :

1. Achieving precise and repeatable high-current test conditions while **controlling R/XL combinations**.
2. Ensuring **safe fault generation** and **rapid isolation** during high-energy short-circuit events.
3. Accurately capturing **high-speed voltage, current, waveform**, and MCB response-time data.

Economical Feasibility :

1. **Lower Cost** : Uses affordable, reusable components and simple circuitry.
2. **Lower Labour** : Automation reduces manual testing and operator effort.
3. **Lower Operating Cost** : Reduces testing time and resource usage.

SDG 9 – Promoting innovation in safer and smarter electrical infrastructure.



SDG 7 - Supporting safer and more reliable electrical energy systems.

Strategies :

1. **Automate** : **R/XL selection** and fault generation
2. **Monitor** : voltage, current, waveform, and trip response in real time.
3. **Ensure safety** : through **automatic fault isolation** and **protection mechanisms**.

Direct Beneficiaries who are Impacting :

1. **MCB Manufacturers** : Faster and more consistent product testing.
2. **Testing & Certification Laboratories** : Automated testing and reliable data collection
3. **Electrical Safety Authorities** : Improved verification and safety assurance.

Strategic Impact :

1. **Sustainability** : **Reusable components** and controlled testing **reduce material and energy wastage**.
2. **Efficiency** : Automation enables **faster, consistent, and repeatable testing**.
3. **Safety** : **Automatic fault detection** and **shutdown** reduce operator exposure.

Strategic Benefits :

1. **Service Delivery** : Faster, reliable, and consistent MCB testing.
2. **Long-Term Impact** : Scalable platform for future testing and laboratory applications.
3. **Technology Transfer** : Supports training, research, and laboratory development in automated testing.

Economical Impact :

1. **Accessibility** : Uses readily available, affordable components for easy adoption.
2. **Employment** : Creates opportunities in automation, testing, and electrical system development.
3. **Cost Benefit** : Reduces equipment, labour, and testing costs through automation.

REFERENCE :

IEEE – Design and Construction of an Automated Test Bench for MCB Testing

<https://ieeexplore.ieee.org/document/7480150>

Times of India – CEA urges power utilities to use IoT technology to prevent short-circuit fires

<https://timesofindia.indiatimes.com/india/cea-urges-power-utilities-to-use-iot-tech-to-prevent-fires-caused-by-short-circuits/articleshow/132078151.cms>

IEC 60898-1:2015 — Short-circuit test section reference

<https://webstore.iec.ch/en/publication/21972>

High-Performance Breaking and Intelligent of Miniature Circuit Breakers

<https://www.semanticscholar.org/paper/High-Performance-Breaking-and-Intelligent-of-Yin-Lang/870529d8d48067bc4c9da7f55cc78739b85c1be3>

Automated High-Current Short Circuit Test System for MCB

<https://www.sihbuddy.in/ps/SIH26029>

Circuit-Breaker Automated Failure Tracking Based on Coil Current Signature

<https://ieeexplore.ieee.org/document/6587111>

IEEE Guide for Power System Protection Testing (C37.233-2023)

<https://ieeexplore.ieee.org/document/10258048>